

Qn	Suggested Solutions
7(a)	Binding energy per nucleon is a maximum/close to the maximum at around $A = 56$.
	Products of splitting a ${}^{56}_{26}\text{Fe}$ nucleus have a lower total binding energy which requires a net input of energy
(b)	energy released = BE of products – BE of reactants $= (1.273 + 0.8405 - 1.910)(1.66 \times 10^{-27})(3.00 \times 10^8)^2$
	$= 3.04 \times 10^{-11} \text{ J}$
(c)	$\lambda = \frac{\ln 2}{28.8 \text{ yrs}}$ $N = N_0 e^{-\lambda t}$ $\frac{N}{N_0} = e^{-\frac{\ln 2}{28.8}(144)}$ $= \frac{1}{32}$ $R = \frac{N_0 - N}{N}$ $= \frac{N_0}{N} - 1$ $= 32 - 1$ $= 31$ OR $\text{no. of half-lives} = \frac{144}{28.8} = 5.00$ $N = \frac{N_0}{2^5}$ $= \frac{1}{32} N_0$ $R = \frac{N_0 - N}{N}$ $= \frac{N_0}{N} - 1$ $= 32 - 1$ $= 31$
(d)(i)	$N_0 = \frac{0.13}{90 \times 1.66 \times 10^{-27}}$ $= 8.7 \times 10^{23}$
(ii)	$A_0 = \lambda N_0$ $= \left(\frac{\ln 2}{28.8 \times 365 \times 24 \times 60 \times 60} \right) (8.70 \times 10^{23})$ $= 6.64 \times 10^{14}$ $\approx 6.6 \times 10^{14} \text{ Bq}$
(iii)	$P = A \times \text{energy released per decay}$ $= (6.64 \times 10^{14})(0.546 \times 10^6 \times 1.60 \times 10^{-19})$
	$P = 58 \text{ W}$

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H2 (9749) Physics

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